5.4 The Remez Algorithm

In this section, we discuss the *Remez*[9] *algorithm* [59, 60], an iterative method to numerically compute the (strongly unique) best approximation $s^* \in \mathcal{S}$ to $f \in \mathscr{C}[a,b] \setminus \mathcal{S}$, where $[a,b] \subset \mathbb{R}$ is a compact interval. Moreover, $\mathcal{S} \subset \mathscr{C}[a,b]$ denotes a Haar space of dimension $n \in \mathbb{N}$ on $[a,b]$.

In any of its iteration steps, the Remez algorithm computes, for an ordered (i.e., monotonically increasing) *reference set* $X = (x_1, \ldots, x_{n+1}) \in [a,b]^{n+1}$ of length $|X| = n+1$, the corresponding (strongly unique) best approximation s_X^* to f with respect to $\| \cdot \|_{\infty,X}$, so that

$$\|s_X^* - f\|_{\infty,X} < \|s - f\|_{\infty,X} \qquad \text{for all } s \in \mathcal{S} \setminus \{s_X^*\}.$$

To compute s_X^*, we first fix an ordered basis $\mathcal{H} = (s_1, \ldots, s_n)$ of the Haar space $\mathcal{S}$, so that s_X^* can be represented as linear combination

$$s_X^* = \sum_{j=1}^{n} \alpha_j^* s_j \in \mathcal{S} \tag{5.30}$$

of the Haar system $\mathcal{H}$ with coefficients $\alpha^* = (\alpha_1^*, \ldots, \alpha_n^*)^T \in \mathbb{R}^n$. According to the alternation theorem, Theorem 5.34, the sought best approximation s_X^* necessarily satisfies the *alternation condition*

$$(s_X^* - f)(x_k) = (-1)^{k-1} \sigma \|s_X^* - f\|_{\infty,X} \qquad \text{for } 1 \le k \le n+1 \tag{5.31}$$

for some $\sigma \in \{\pm 1\}$. Letting $\eta_X = \sigma \|s_X^* - f\|_{\infty,X}$ we rewrite (5.31) as

$$s_X^*(x_k) + (-1)^k \eta_X = f(x_k) \qquad \text{for } 1 \le k \le n+1. \tag{5.32}$$

Therefore, η_X and the unknown coefficients $\alpha^* \in \mathbb{R}^n$ of s_X^* are the solution of the linear equation system

$$A_{\varepsilon,\mathcal{H},X}^T \cdot \begin{bmatrix} \eta_X \\ \alpha^* \end{bmatrix} = f_X \tag{5.33}$$

with the right hand side $f_X = (f(x_1), \ldots, f(x_{n+1}))^T \in \mathbb{R}^{n+1}$ and the alternation matrix $A_{\varepsilon,\mathcal{H},X} \in \mathbb{R}^{(n+1)\times(n+1)}$ in (5.27), containing the sign vector $\varepsilon = (-1, 1, \ldots, (-1)^{n+1}) \in \{\pm 1\}^{n+1}$, or,

$$\left[\begin{array}{c|ccc} -1 & s_1(x_1) & \cdots & s_n(x_1) \\ 1 & s_1(x_2) & \cdots & s_n(x_2) \\ \vdots & \vdots & & \vdots \\ (-1)^{n+1} & s_1(x_{n+1}) & \cdots & s_n(x_{n+1}) \end{array}\right] \left[\begin{array}{c} \eta_X \\ \hline \alpha_1^* \\ \vdots \\ \alpha_n^* \end{array}\right] = \begin{bmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_{n+1}) \end{bmatrix}.$$

By Proposition 5.32 (b), the matrix $A_{\varepsilon,\mathcal{H},X}$ is non-singular, and so the solution of the linear system (5.33) is unique. By the solution of (5.33), we

[9] Evgeny Yakovlevich Remez (1896-1975), mathematician

do not only obtain the coefficients $\alpha^* = (\alpha_1^*, \ldots, \alpha_n^*)^T \in \mathbb{R}^n$ of the best approximation s_X^* in (5.30), but also by $|\eta_X| = \|s_X^* - f\|_{\infty,X}$ we get the minimal distance and the sign $\sigma = \operatorname{sgn}(\eta_X)$ in (5.31).

The following observation concerning Chebyshev approximation to a function $f \in \mathscr{C}[a,b]$ by algebraic polynomials from $\mathcal{P}_{n-1}$ shows that, in this special case, the linear system (5.33) can be avoided. To this end, we use the Newton representation (2.34) for the interpolation polynomial in Theorem 2.13. Recall that the *Newton polynomials* are given as

$$\omega_k(x) = \prod_{j=1}^{k} (x - x_j) \in \mathcal{P}_k \qquad \text{for } 0 \le k \le n-1.$$

In particular, we apply the linear operator $[x_1, \ldots, x_{n+1}] : \mathscr{C}[a,b] \longrightarrow \mathbb{R}$ of the *divided differences* to f (see Definition 2.10). To evaluate $[x_1, \ldots, x_{n+1}](f)$, we apply the recursion in Theorem 2.14. The recursion in Theorem 2.14 operates only on the vector of function values $f_X = (f(x_1), \ldots, f(x_{n+1}))^T \in \mathbb{R}^{n+1}$. Therefore, the application of $[x_1, \ldots, x_{n+1}]$ is also well-defined for any sign vector $\varepsilon \in \{\pm 1\}^{n+1}$ of length $n+1$. In particular, the divided difference $[x_1, \ldots, x_{n+1}](\varepsilon)$ can also be evaluated by the recursion in Theorem 2.14. In the formulation of the following result, we apply divided differences to vectors ε with alternating signs.

Proposition 5.35. *For $n \in \mathbb{N}$, let $X = (x_1, \ldots, x_{n+1})$ be an ordered set of $n+1$ points in $[a,b] \subset \mathbb{R}$, $\varepsilon = (-1, 1, \ldots, (-1)^{n+1}) \in \{\pm 1\}^{n+1}$ a sign vector, and $f \in \mathscr{C}[a,b] \setminus \mathcal{P}_{n-1}$. Then,*

$$s_X^* = \sum_{k=0}^{n-1} [x_1, \ldots, x_{k+1}](f - \eta_X \varepsilon)\omega_k \in \mathcal{P}_{n-1} \tag{5.34}$$

is the strongly unique best approximation $s_X^* \in \mathcal{P}_{n-1}$ to f w.r.t. $\|\cdot\|_{\infty,X}$, where

$$\eta_X = \frac{[x_1, \ldots, x_{n+1}](f)}{[x_1, \ldots, x_{n+1}](\varepsilon)}. \tag{5.35}$$

The minimal distance is given as

$$\|s_X^* - f\|_{\infty,X} = |\eta_X|.$$

Proof. Application of the linear operator $[x_1, \ldots, x_{n+1}] : \mathscr{C}[a,b] \longrightarrow \mathbb{R}$ to the alternation condition (5.32) immediately gives the representation

$$\eta_X = \frac{[x_1, \ldots, x_{n+1}](f)}{[x_1, \ldots, x_{n+1}](\varepsilon)}.$$

Indeed, due to Corollary 2.18 (b), all polynomials from $\mathcal{P}_{n-1}$ are contained in the kernel of $[x_1, \ldots, x_{n+1}]$. In particular, we have $[x_1, \ldots, x_{n+1}](s_X^*) = 0$.

Under the alternation condition (5.32), $s_X^* \in \mathcal{P}_{n-1}$ is the unique solution of the interpolation problem

$$s_X^*(x_k) = f(x_k) - (-1)^k \eta_X \qquad \text{for } 1 \le k \le n,$$

already for the first n alternation points $(x_1, \dots, x_n) \in E^n_{s_X^* - f}$. This gives the stated Newton representation of s_X^* in (5.34). ■

Remark 5.36. Note that all divided differences

$$[x_1, \dots, x_{k+1}](f - \eta_X \varepsilon) = [x_1, \dots, x_{k+1}](f) - \eta_X [x_1, \dots, x_{k+1}](\varepsilon)$$

in (5.34) are readily available from the computation of η_X in (5.35). Therefore, we can compute the best approximation $s_X^* \in \mathcal{P}_{n-1}$ to f with respect to $\|\cdot\|_{\infty,X}$ by divided differences in only $\mathcal{O}(n^2)$ steps, where the computation is efficient and stable. □

To show how the result of Proposition 5.35, in combination with Remark 5.36, can be applied, we make the following concrete example.

Example 5.37. Let $\mathcal{F} = \mathscr{C}[0,2]$ and $\mathcal{S} = \mathcal{P}_1 \subset \mathcal{F}$. We approximate the exponential function $f(x) = \mathrm{e}^x$ on the reference set $X = \{0,1,2\}$. To compute the minimal distance η_X and the best approximation $s_X^* \in \mathcal{P}_1$ we use Proposition 5.35 with $n = 2$. We apply divided differences to the sign vector $\varepsilon = (-1, 1, -1)$ and to the data vector $f_X = (1, e, e^2)$, where e denotes the *Euler number*. By the recursion in Theorem 2.14, we obtain the following triangular scheme for divided differences (see Table 2.1).

X	f_X		
0	1		
1	e	$e-1$	
2	e^2	$e(e-1)$	$(e-1)^2/2$

X	ε_X		
0	-1		
1	1	2	
2	-1	-2	-2

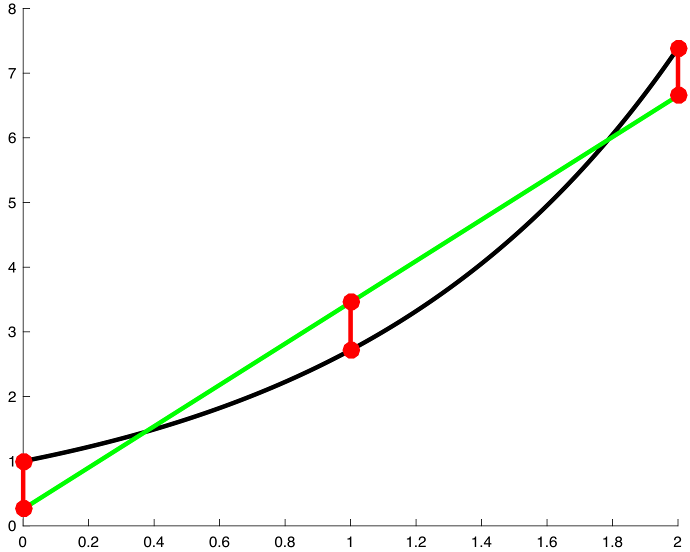
Hereby we obtain

$$\eta_X = \frac{[0,1,2](f)}{[0,1,2](\varepsilon)} = -\left(\frac{e-1}{2}\right)^2 \qquad \text{and so} \qquad \|s_X^* - f\|_{\infty,X} = \left(\frac{e-1}{2}\right)^2.$$

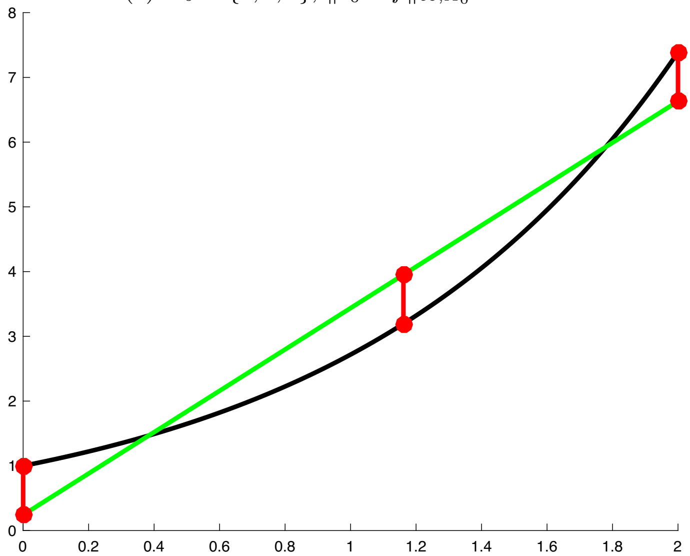
Moreover,

$$s_X^* = [0](f - \eta_X \varepsilon) + [0,1](f - \eta_X \varepsilon)x = 1 - \left(\frac{e-1}{2}\right)^2 + \frac{e^2-1}{2}x$$

is the unique best approximation to f from $\mathcal{P}_1$ w.r.t. $\|\cdot\|_{\infty,X}$ (see Fig. 5.5 (a)). ♢



(a) $X_0 = \{0, 1, 2\}$, $\|s_0^* - f\|_{\infty, X_0} \approx 0.7381$



(b) $X_1 = \{0, x^*, 2\}$, $\|s_1^* - f\|_{\infty, X_1} \approx 0.7579$

Fig. 5.5. Approximation to $f(x) = \mathrm{e}^x$ on $[0, 2]$ by linear polynomials from $\mathcal{P}_1$. (a) Initial reference set $X_0 = \{0, 1, 2\}$ with minimal distance $|\eta_0| = (e-1)^2/4 \approx 0.7381$. (b) Reference set $X_1 = \{0, x^*, 2\}$, where $x^* = \log((e^2 - 1)/2) \approx 1.1614$, with minimal distance $|\eta_1| = \frac{1}{4}\left[(e^2 - 1)(x^* - 1) + 2\right] \approx 0.7579$ (see Example 5.45).

We present another example, where we link with Example 5.7.

Example 5.38. We approximate the absolute-value function $f(x) = |x|$ on $[-1, 1]$ by quadratic polynomials, i.e., $\mathcal{S} = \mathcal{P}_2$. By our previous investigations in Example 5.7, $E_{p_2^*-f} = \{-1, -1/2, 0, 1/2, 1\}$ is the set of extremal points for the best approximation $p_2^* \in \mathcal{P}_2$ to f. To compute p_2^*, we apply Proposition 5.35 with $n = 3$. We let $\varepsilon = (-1, 1, -1, 1)$ and we choose $X = \{-1, -1/2, 0, 1/2\} \subset E_{p_2^*-f}$ as the reference set. By the recursion in Theorem 2.14, we obtain the following triangular scheme (see Table 2.1).

X	f_X			
-1	1			
$-\frac{1}{2}$	$\frac{1}{2}$	-1		
0	0	-1	0	
$\frac{1}{2}$	$\frac{1}{2}$	1	2	$\frac{4}{3}$

X	ε_X			
-1	-1			
$-\frac{1}{2}$	1	4		
0	-1	-4	-8	
$\frac{1}{2}$	1	4	8	$\frac{32}{3}$

Hereby we obtain $\eta_X = 1/8$, and so $\|s_X^* - f\|_{\infty,X} = 1/8$, along with

$$[x_1](f - \eta_X \varepsilon) = \frac{9}{8}, \quad [x_1, x_2](f - \eta_X \varepsilon) = -\frac{3}{2}, \quad [x_1, x_2, x_3](f - \eta_X \varepsilon) = 1.$$

Therefore,

$$s_X^*(x) = \frac{9}{8} - \frac{3}{2}(x+1) + (x+1)\left(x + \frac{1}{2}\right) = \frac{1}{8} + x^2$$

is the unique best approximation to f from $\mathcal{P}_2$ with respect to $\|\cdot\|_\infty$. Note that this is consistent with our observations in Example 5.7, since $p_2^* \equiv s_X^*$. $\diamondsuit$

Now we describe the iteration steps of the Remez algorithm. At any *Remez step* the current reference set (in increasing order)

$$X = (x_1, \ldots, x_{n+1}) \in [a, b]^{n+1}$$

is modified. This is done by a *Remez exchange* of one point $\hat{x} \in X$ for one point $x^* \in [a, b] \setminus X$, where

$$|(s_X^* - f)(x^*)| = \|s_X^* - f\|_\infty,$$

so that the next reference set is

$$X_+ = (X \setminus \{\hat{x}\}) \cup \{x^*\} = (x_1^+, \ldots, x_{n+1}^+) \in [a, b]^{n+1}.$$

With the Remez exchange, the point x^* is swapped for the point $\hat{x} \in X$, such that the points of the new reference set X_+ are in increasing order, i.e.,

$$a \le x_1^+ < x_2^+ < \ldots < x_n^+ < x_{n+1}^+ \le b,$$

and with maintaining the alternation condition, i.e.,

$$\operatorname{sgn}((s_X^* - f)(x_j^+)) = (-1)^j \sigma \qquad \text{for } 1 \le j \le n+1$$

for some $\sigma \in \{\pm 1\}$. The exchange for the point pair $(\hat{x}, x^*) \in X \times [a,b] \setminus X$ is described by the *Remez exchange*, Algorithm 8.

Algorithm 8 Remez exchange

```
function REMEZ EXCHANGE(X, s*_X)
    Input: reference set X = (x_1, ..., x_{n+1}) ∈ [a,b]^{n+1};
           best approximation s*_X to f with respect to ‖·‖_{∞,X};

    find x* ∈ [a,b] satisfying |(s*_X − f)(x*)| = ‖s*_X − f‖_∞;
    let σ* := sgn((s*_X − f)(x*);

    if x* ∈ X then return X;                          ▷ best approximation found
    else if x* < x_1 then
        if sgn((s*_X − f)(x_1)) = σ* then X_+ = (x*, x_2, ..., x_{n+1});
        else X_+ = (x*, x_1, ..., x_n);
        end if
    else if x* > x_{n+1} then
        if sgn((s*_X − f)(x_{n+1})) = σ* then X_+ = (x_1, ..., x_n, x*);
        else X_+ = (x_2, ..., x_{n+1}, x*);
        end if
    else
        find j ∈ {1, ..., n} satisfying x_j < x* < x_{j+1};
        if sgn((s*_X − f)(x_j)) = σ* then X_+ = (x_1, ..., x_{j−1}, x*, x_{j+1}, ..., x_{n+1});
        else X_+ = (x_1, ..., x_j, x*, x_{j+2}, ..., x_{n+1});
        end if
    end if
    return X_+;
end function
```

Remark 5.39. The reference set $X_+ = (x_1^+, \ldots, x_{n+1}^+) \in [a,b]^{n+1}$, after the application of one Remez exchange, Algorithm 8, to the previous reference set $X = (x_1, \ldots, x_{n+1}) \in [a,b]^{n+1}$, satisfies the following three conditions.

- $|(s_X^* - f)(x^*)| = \|s_X^* - f\|_\infty$ for one $x^* \in X_+$;
- $|(s_X^* - f)(x)| \ge \|s_X^* - f\|_{\infty,X}$ for all $x \in X_+$;
- $\operatorname{sgn}((s_X^* - f)(x_j^+)) = (-1)^j \sigma$ for $1 \le j \le n+1$ and some $\sigma \in \{\pm 1\}$;

These conditions are required for the performance of the Remez algorithm. □

Now we formulate the Remez algorithm, Algorithm 9, as an iterative method to numerically compute the (strongly unique) best approximation $s^* \in \mathcal{S}$ to $f \in \mathscr{C}[a,b] \setminus \mathcal{S}$ satisfying

$$\eta = \|s^* - f\|_\infty < \|s - f\|_\infty \qquad \text{for all } s \in \mathcal{S} \setminus \{s^*\}.$$

The Remez algorithm generates a sequence $(X_k)_{k \in \mathbb{N}_0} \subset [a,b]^{n+1}$ of reference sets, so that for the transition from $X = X_k$ to $X_+ = X_{k+1}$, for any $k \in \mathbb{N}_0$, all three conditions in Remark 5.39 are satisfied. The corresponding sequence of best approximations $s_k^* \in \mathcal{S}$ to f with respect to $\|\cdot\|_{\infty,X_k}$ satisfying

$$\eta_k = \|s_k^* - f\|_{\infty,X_k} < \|s - f\|_{\infty,X_k} \qquad \text{for all } s \in \mathcal{S} \setminus \{s_k^*\}$$

converges to s^*, i.e., $s_k^* \longrightarrow s^*$ and $\eta_k \longrightarrow \eta$, for $k \to \infty$, as we will prove in Theorem 5.43.

Algorithm 9 Remez algorithm

1: **function** REMEZ ALGORITHM
2:     **Input:** Haar space $\mathcal{S}$ of dimension $n \in \mathbb{N}$; $f \in \mathscr{C}[a,b] \setminus \mathcal{S}$;
3:
4:     **find** initial reference set $X_0 = (x_1^{(0)}, \ldots, x_{n+1}^{(0)}) \in [a,b]^{n+1}$;
5:     **for** $k = 0, 1, 2, \ldots$ **do**
6:         **compute** best approximation $s_k^* \in \mathcal{S}$ to f with respect to $\|\cdot\|_{\infty,X_k}$;
7:         **let** $\eta_k := \|s_k^* - f\|_{\infty,X_k}$;
8:         **compute** $\rho_k = \|s_k^* - f\|_\infty$;
9:         **if** $\rho_k \leq \eta_k$ **then return** s_k^* ▷ best approximation found
10:         **else**
11:             **find** reference set $X_{k+1} = (x_1^{(k+1)}, \ldots, x_{n+1}^{(k+1)}) \in [a,b]^{n+1}$ satisfying
12:             • $|(s_k^* - f)(x^*)| = \rho_k$ for some $x^* \in X_{k+1}$;
13:             • $|(s_k^* - f)(x)| \geq \eta_k$ for all $x \in X_{k+1}$;
14:             • $\operatorname{sgn}((s_k^* - f)(x_j^{(k+1)})) = (-1)^j \sigma_k$
15:               for $1 \leq j \leq n+1$ and some $\sigma_k \in \{\pm 1\}$.   ▷ alternation condition
16:         **end if**
17:     **end for**
18: **end function**

Remark 5.40. We remark that the construction of the reference set X_{k+1} in line 11 of Algorithm 9 can be accomplished by a Remez exchange step, Algorithm 8. In this case, all three conditions in lines 12-15 of Algorithm 9 are satisfied, according to Remark 5.39. □

Next, we analyze the convergence of the Remez algorithm. To this end, we first remark that at any step k in the Remez algorithm, we have

- a current reference set $X_k = (x_1^{(k)}, \ldots, x_{n+1}^{(k)}) \subset [a,b]^{n+1}$,
- *alternating* signs $\varepsilon^{(k)} = (\varepsilon_1^{(k)}, \ldots, \varepsilon_{n+1}^{(k)}) \in \{\pm 1\}^{n+1}$,
- and *positive* coefficients $\lambda^{(k)} = (\lambda_1^{(k)}, \ldots, \lambda_{n+1}^{(k)})^T \in \Lambda_{n+1}$,

so that the dual functional $\varphi : [a,b] \longrightarrow \mathbb{R}$, defined as

$$\varphi(u) = \sum_{j=1}^{n+1} \lambda_j^{(k)} \varepsilon_j^{(k)} u(x_j^{(k)}) \qquad \text{for } u \in \mathscr{C}[a,b],$$

satisfies the characterization (5.29) in the alternation theorem, Theorem 5.34. In particular, by the alternation theorem, the following properties hold.

- $X_k \subset E_{s_k^* - f}$;
- $\varepsilon_j^{(k)} = \operatorname{sgn}((s_k^* - f)(x_j^{(k)})) = (-1)^j \sigma_k$ for all $1 \leq j \leq n+1$ with $\sigma_k \in \{\pm 1\}$;
- $\varepsilon_j^{(k)} (s_k^* - f)(x_j^{(k)}) = \|s_k^* - f\|_{\infty, X_k} = \eta_k$ for all $1 \leq j \leq n+1$;
- $\varphi(s) = 0$ for all $s \in \mathcal{S}$.

Now we prove the monotonicity of the minimal distances η_k.

Proposition 5.41. *Let the assumptions from the Remez algorithm be satisfied. Then, for any step $k \in \mathbb{N}_0$, where the Remez iteration does not terminate, we have the monotonicity of the minimal distances,*

$$\eta_{k+1} > \eta_k.$$

Proof. The representation

$$\begin{aligned}
\eta_{k+1} &= \sum_{j=1}^{n+1} \lambda_j^{(k+1)} \varepsilon_j^{(k+1)} (s_{k+1}^* - f)(x_j^{(k+1)}) \\
&= \sum_{j=1}^{n+1} \lambda_j^{(k+1)} \varepsilon_j^{(k+1)} (s_k^* - f)(x_j^{(k+1)}) \\
&= \sum_{j=1}^{n+1} \lambda_j^{(k+1)} |(s_k^* - f)(x_j^{(k+1)})|
\end{aligned}$$

holds. Moreover, we have

$$\varepsilon_j^{(k+1)} = \operatorname{sgn}(s_k^* - f)(x_j^{(k+1)}) \qquad \text{for all } 1 \leq j \leq n+1$$

after the Remez exchange (see Algorithm 9, line 14).

Moreover, we have $|(s_k^* - f)(x_j^{(k+1)})| \geq \eta_k$, for all $1 \leq j \leq n+1$ (cf. line 13), and there is one index $j^* \in \{1, \ldots, n+1\}$ (cf. line 12) satisfying

$$|(s_k^* - f)(x_{j^*}^{(k+1)})| = \rho_k = \|s_k^* - f\|_\infty.$$

But this implies

$$\eta_{k+1} \geq \lambda_{j^*}^{(k+1)} \rho_k + (1 - \lambda_{j^*}^{(k+1)})\eta_k > \lambda_{j^*}^{(k+1)} \eta_k + (1 - \lambda_{j^*}^{(k+1)})\eta_k = \eta_k, \quad (5.36)$$

which already completes our proof. ■

Next, we show that the coefficients $\lambda_j^{(k)}$ are uniformly bounded away from zero.

Lemma 5.42. *Let $f \in \mathscr{C}[a,b]\backslash\mathcal{S}$. Then, under the assumptions of the Remez algorithm, the uniform bound*

$$\lambda_j^{(k)} \geq \alpha > 0 \qquad \text{for all } 1 \leq j \leq n+1 \text{ and all } k \in \mathbb{N}_0,$$

holds for some $\alpha > 0$ which is independent of $1 \leq j \leq n$ and $k \in \mathbb{N}_0$.

Proof. We have

$$\eta_k = -\sum_{j=1}^{n+1} \lambda_j^{(k)} \varepsilon_j^{(k)} f(x_j^{(k)}) = \sum_{j=1}^{n+1} \lambda_j^{(k)} \varepsilon_j^{(k)} (s_k^* - f)(x_j^{(k)}) \geq \eta_0 = \|s^* - f\|_{\infty, X_0}.$$

Suppose the statement is false. Then, there are sequences of reference sets $(X_k)_k$, signs $(\varepsilon^{(k)})_k$, and coefficients $(\lambda^{(k)})_k$ satisfying

$$\eta_k = -\sum_{j=1}^{n+1} \lambda_j^{(k)} \varepsilon_j^{(k)} f(x_j^{(k)}) \geq \eta_0 > 0 \qquad \text{for all } k \in \mathbb{N}_0, \quad (5.37)$$

where one index $j^* \in \{1, \ldots, n+1\}$ satisfies $\lambda_{j^*}^{(k)} \longrightarrow 0$, for $k \to \infty$.

But the elements of the sequences $(X_k)_k$, $(\varepsilon^{(k)})_k$, and $(\lambda^{(k)})_k$ lie in compact sets, respectively. Therefore, there are convergent subsequences with

$$\begin{aligned} x_j^{(k_\ell)} &\longrightarrow x_j \in [a,b] \quad \text{for } \ell \to \infty, \\ \varepsilon_j^{(k_\ell)} &\longrightarrow \varepsilon_j \in \{\pm 1\} \quad \text{for } \ell \to \infty, \\ \lambda_j^{(k_\ell)} &\longrightarrow \lambda_j \in [0,1] \quad \text{for } \ell \to \infty, \end{aligned}$$

for all $1 \leq j \leq n+1$, where $\lambda_{j^*} = 0$ for one index $j^* \in \{1, \ldots, n+1\}$.

Now we regard an interpolant $s \in \mathcal{S}$ satisfying $s(x_j) = f(x_j)$ for all $1 \leq j \leq n+1$, $j \neq j^*$. Then, we have

$$\begin{aligned} \eta_{k_\ell} &= \sum_{j=1}^{n+1} \lambda_j^{(k_\ell)} \varepsilon_j^{(k_\ell)} (s_{k_\ell}^* - f)(x_j^{(k_\ell)}) = \sum_{j=1}^{n+1} \lambda_j^{(k_\ell)} \varepsilon_j^{(k_\ell)} (s - f)(x_j^{(k_\ell)}) \\ &= \sum_{\substack{j=1 \\ j \neq j^*}}^{n+1} \lambda_j^{(k_\ell)} \varepsilon_j^{(k_\ell)} (s - f)(x_j^{(k_\ell)}) + \lambda_{j^*}^{(k_\ell)} \varepsilon_{j^*}^{(k_\ell)} (s - f)(x_{j^*}^{(k_\ell)}) \\ &\longrightarrow \sum_{\substack{j=1 \\ j \neq j^*}}^{n+1} \lambda_j \varepsilon_j (s - f)(x_j) + \lambda_{j^*} \varepsilon_{j^*} (s - f)(x_{j^*}) = 0 \quad \text{for } \ell \to \infty. \end{aligned}$$

But this is in contradiction to (5.37). ■

Now we can finally prove convergence for the Remez agorithm.

Theorem 5.43. *Either the Remez algorithm, Algorithm 9, terminates after $k \in \mathbb{N}$ steps with returning the best approximation $s_k^* = s^*$ to $f \in \mathscr{C}[a,b]$ or the Remez algorithm generates convergent sequences of minimal distances $(\eta_k)_k$ and best approximations $(s_k^*)_k$ with limit elements*

$$\lim_{k\to\infty} \eta_k = \eta = \|s^* - f\|_\infty \quad \text{and} \quad \lim_{k\to\infty} s_k^* = s^* \in \mathcal{S},$$

where $s^* \in \mathcal{S}$ is the strongly unique best approximation to $f \in \mathscr{C}[a,b]$ with minimal distance η. The sequence $(\eta_k)_k$ of minimal distances converges linearly to η by the contraction

$$\eta - \eta_{k+1} < \theta(\eta - \eta_k) \qquad \text{for some } \theta \in (0,1). \tag{5.38}$$

Proof. Let $f \in \mathscr{C}[a,b] \setminus \mathcal{S}$ (for $f \in \mathcal{S}$ the statement is trivial).

If the Remez algorithm terminates after $k \in \mathbb{N}$ steps, in line 9 of Algorithm 9, with $s_k^* \in \mathcal{S}$, then $s_k^* = s^*$ is, according to the alternation theorem, Theorem 5.34, the strongly unique best approximation to f.

Now suppose the Remez algorithm does not terminate after finitely many steps. For this case, we first show the contraction property (5.38).

By the estimate in (5.36),

$$\eta_{k+1} \geq \lambda_{j^*}^{(k+1)} \rho_k + (1 - \lambda_{j^*}^{(k+1)})\eta_k \tag{5.39}$$

and from $\rho_k = \|s_k^* - f\|_\infty > \|s^* - f\|_\infty = \eta > 0$ it follows that

$$\eta_{k+1} > \lambda_{j^*}^{(k+1)} \eta + (1 - \lambda_{j^*}^{(k+1)})\eta_k$$

and so

$$\eta - \eta_{k+1} < (1 - \lambda_{j^*}^{(k+1)})(\eta - \eta_k).$$

By Lemma 5.42, there is one $\alpha > 0$ satisfying $\lambda_j^{(k+1)} \geq \alpha$, for all $1 \leq j \leq n+1$ and all $k \in \mathbb{N}_0$. Therefore, the stated contraction (5.38) holds for $\theta = 1 - \alpha \in (0,1)$. From this, we get the estimate

$$\eta - \eta_k < \theta^k(\eta - \eta_0) \qquad \text{for all } k \in \mathbb{N}_0$$

by induction on k. Therefore, the sequence $(\eta_k)_k$ of minimal distances is convergent with limit element η, i.e., $\eta_k \longrightarrow \eta$, for $k \to \infty$.

From estimate (5.39), we can conclude

$$\rho_k \leq \frac{\eta_{k+1} - \eta_k}{\lambda_{j^*}^{(k+1)}} + \eta_k < \frac{\eta_{k+1} - \eta_k}{1 - \theta} + \eta_k,$$

and this gives the estimates

$$\eta_k < \rho_k < \frac{\eta_{k+1} - \eta_k}{1 - \theta} + \eta_k.$$

This implies the convergence of the distances ρ_k to η, i.e.,

$$\lim_{k \to \infty} \rho_k = \lim_{k \to \infty} \|s_k^* - f\|_\infty = \|s^* - f\|_\infty = \eta.$$

We can conclude that the sequence $(s_k^*)_k \subset \mathcal{S}$ of the strongly unique best approximations to f on X_k converges to the strongly unique best approximation s^* to f. ■

Finally, we discuss one important observation. We note that for the approximation of *strictly convex* functions $f \in \mathscr{C}[a,b]$ by linear polynomials, the Remez algorithm may return the best approximation $s^* \in \mathcal{P}_1$ to f after only one step.

Proposition 5.44. *Let $f \in \mathscr{C}[a,b]$ be a strictly convex function on a compact interval $[a,b]$ and $\mathcal{S} = \mathcal{P}_1$. Moreover, let $X_0 = (a, x_0, b)$, for $x_0 \in (a,b)$, be an initial reference set for the Remez algorithm. Then, the Remez algorithm terminates after at most one Remez exchange.*

Proof. Regard $s \in \mathcal{P}_1$ in its monomial representation $s(x) = m \cdot x + c$ for $m, c \in \mathbb{R}$. Then, we have for $x, y \in [a,b]$, $x \neq y$, and $\lambda \in (0,1)$ the strict inequality

$$\begin{aligned}
&(f - s)(\lambda x + (1-\lambda)y) \\
&= f(\lambda x + (1-\lambda)y) - m \cdot (\lambda x + (1-\lambda)y) - c \\
&< \lambda f(x) + (1-\lambda) f(y) - m \cdot (\lambda x + (1-\lambda)y) - c \\
&= \lambda f(x) - \lambda m x - \lambda c + (1-\lambda) f(y) - (1-\lambda) m y - (1-\lambda) c \\
&= \lambda (f - s)(x) + (1-\lambda)(f - s)(y)
\end{aligned}$$

by the strict convexity of f, i.e., $f - s$ is also strictly convex.

Now let $s^* \in \mathcal{P}_1$ be the strongly unique best approximation to f. Due to the alternation theorem, Theorem 5.34, the error function $f - s^*$ has at least *three* extremal points with alternating signs in $[a,b]$. Since $f - s^*$ is strictly convex and continuous, $f - s^*$ has exactly one global minimum x^* on (a,b). Moreover, two global maxima of $f - s^*$ are at the boundary of $[a,b]$, i.e., we have $\{a, b\} \subset E_{s^* - f}$ with

$$(f - s^*)(a) = \|f - s^*\|_\infty = (f - s^*)(b).$$

From the representation $s^*(x) = m^* \cdot x + c^*$, we obtain the slope

$$m^* = \frac{f(b) - f(a)}{b - a} = [a,b](f).$$

Let $s_0^* \in \mathcal{P}_1$ be the best approximation to f with respect to $X_0 = (a, x_0, b)$. Then, according to the alternation theorem, we have

$$(f - s_0^*)(a) = \sigma\|f - s_0^*\|_{\infty,X_0} = (f - s_0^*)(b) \qquad \text{for some } \sigma \in \{\pm 1\},$$

whereby the representation $s_0^*(x) = m_0 \cdot x + c_0$ implies $m_0 = [a, b](f) = m^*$, i.e., s^* and s_0^* differ by at most one constant.

If $x_0 \in E_{s^*-f}$, then $x_0 = x^*$, and the best approximation s^* to f is already found by s_0^*, due to the alternation theorem. In this case, the Remez algorithm terminates immediately with returning $s^* = s_0^*$.

If $x_0 \notin E_{s^*-f}$, then the Remez algorithm selects the unique global minimum $x^* \neq x_0$ of $f - s^*$ for the exchange with x_0: Since s^* and s_0^* differ by at most a constant, x^* is also the unique global minimum of $f - s_0^*$ on (a, b), i.e., we have

$$(f - s_0^*)(x^*) < (f - s_0^*)(x) \qquad \text{for all } x \in [a, b], \text{ where } x \neq x^*. \tag{5.40}$$

By the strict convexity of $f - s_0^*$, we can further conclude the strict inequality

$$(f - s_0^*)(x^*) < (f - s_0^*)(x_0) < 0,$$

or,

$$\rho_0 = \|f - s_0^*\|_\infty = |(f - s_0^*)(x^*)| > |(f - s_0^*)(x_0)| = \|f - s_0^*\|_{\infty,X_0} = \eta_0. \tag{5.41}$$

By (5.40) and (5.41), the point x^* is the unique global maximum of $|f - s_0^*|$ on $[a, b]$. Therefore, x^* is the only candidate for the required Remez exchange (in line 5 of Algorithm 8) for x_0. After the execution of the Remez exchange, we have $X_1 = (a, x^*, b)$, so that the Remez algorithm immediately terminates with returning $s_1^* = s^*$. ■

For further illustration, we make an example linked with Example 5.37.

Example 5.45. We approximate the strictly convex exponential function $f(x) = \exp(x)$ on the interval $[0, 2]$ by linear polynomials, i.e., $\mathcal{F} = \mathscr{C}[0, 2]$ and $\mathcal{S} = \mathcal{P}_1$. We take $X_0 = (0, 1, 2)$ as the initial reference set in the Remez algorithm, Algorithm 9. According to Example 5.37,

$$s_0^*(x) = 1 - \left(\frac{e-1}{2}\right)^2 + \frac{e^2-1}{2}x$$

is the unique best approximation to f from $\mathcal{P}_1$ with respect to $\|\cdot\|_{\infty,X_0}$, with minimal distance $|\eta_0| = (e-1)^2/4 \approx 0.7381$, where e is the Euler number.

The error function $|s_0^*(x) - \exp(x)|$ attains on $[0, 2]$ its unique maximum $\rho_0 = \|s_0^* - \exp\|_\infty \approx 0.7776$ at $x^* = \log((e^2 - 1)/2) > 1$. We have $\rho_0 > \eta_0$, and so one Remez exchange leads to the new reference set $X_1 = (0, x^*, 2)$.

According to Proposition 5.44, the Remez algorithm returns already after the next iteration the best approximation s_1^* to f.

Finally, we compute s_1^*, the best approximation to f for the reference set $X_1 = (0, x^*, 2)$. To this end, we proceed as in Example 5.37, where we first determine the required divided differences for f and $\varepsilon = (-1, 1, -1)$ by using the recursion in Theorem 2.14:

X	f_X		
0	1		
x^*	$\frac{e^2-1}{2}$	$\frac{e^2-3}{2x^*}$	
2	e^2	$\frac{e^2+1}{2(2-x^*)}$	$\frac{(e^2-1)(x^*-1)+2}{2(2-x^*)x^*}$

X	ε_X		
0	-1		
x^*	1	$\frac{2}{x^*}$	
2	-1	$-\frac{2}{2-x^*}$	$-\frac{2}{(2-x^*)x^*}$

From this we compute the minimal distance $\|s_1^* - f\|_{\infty,X_1} = -\eta_1 \approx 0.7579$ by

$$\eta_1 = -\frac{1}{4}\left[(e^2-1)(x^*-1)+2\right]$$

and the best approximation to f from $\mathcal{P}_1$ with respect to $\|\cdot\|_{\infty,X_1}$ by

$$s_1^* = [0](f - \eta_X\varepsilon) + [0, x^*](f - \eta_X\varepsilon)x = 1 + \eta_1 + \frac{e^2 - 4\eta_1 - 3}{2x^*}x.$$

By Proposition 5.44, the Remez algorithm terminates with the reference set $X_1 = E_{s_1^*-f}$, so that by $s_1^* \in \mathcal{P}_1$ the unique best approximation to f with respect to $\|\cdot\|_\infty$ is found. Figure 5.5 shows the best approximations $s_j^* \in \mathcal{P}_1$ to f for the reference sets X_j, for $j = 0, 1$. $\diamondsuit$

5.5 Exercises

Exercise 5.46. Let $\mathcal{F} = \mathscr{C}[-1,1]$ be equipped with the maximum norm $\|\cdot\|_\infty$. Moreover, let $f \in \mathcal{P}_3 \setminus \mathcal{P}_2$ be a cubic polynomial, i.e., has the form

$$f(x) = a\,x^3 + b\,x^2 + c\,x + d \qquad \text{for } x \in [-1,1]$$

with coefficients $a, b, c, d \in \mathbb{R}$, where $a \neq 0$.

(a) Compute a best approximation $p_2^* \in \mathcal{P}_2$ to f from $\mathcal{P}_2$ w.r.t. $\|\cdot\|_\infty$.
(b) Is the best approximation p_2^* from (a) unique?

Exercise 5.47. Let $P_\infty : \mathscr{C}[a,b] \longrightarrow \mathcal{P}_n$ denote the operator, which assigns every $f \in \mathscr{C}[a,b]$ to its best approximation $p_\infty^*(f) \in \mathcal{P}_n$ from $\mathcal{P}_n$ w.r.t. $\|\cdot\|_\infty$, i.e.,

$$P_\infty(f) = p_\infty^*(f) \qquad \text{for } f \in \mathscr{C}[a,b].$$

(a) Show that P_∞ is well-defined.
(b) Is P_∞ linear or non-linear?

Exercise 5.48. For a compact interval $[a,b] \subset \mathbb{R}$, let $\mathcal{F} = \mathscr{C}[a,b]$ be equipped with the maximum norm $\|\cdot\|_\infty$. Moreover, let $f \in \mathscr{C}[a,b] \setminus \mathcal{P}_{n-1}$, for $n \in \mathbb{N}$. Then, there is a strongly unique best approximation $p^* \in \mathcal{P}_{n-1}$ to f from $\mathcal{P}_{n-1}$ w.r.t. $\|\cdot\|_\infty$ and an alternation set $X = (x_1, \ldots, x_{n+1}) \in E_{s^*-f}^{n+1}$ for s^* and f (see Corollary 5.4). For the dual characterization of the best approximation $p^* \in \mathcal{P}_{n-1}$ we use, as in (5.6), a linear functional $\varphi \in \mathcal{F}'$ of the form

$$\varphi(u) = \sum_{k=1}^{n+1} \lambda_k \, \varepsilon_k \, u(x_k) \qquad \text{for } u \in \mathscr{C}[a,b]$$

with coefficients $\lambda = (\lambda_1, \ldots, \lambda_{n+1})^T \in \Lambda_{n+1}$ and alternating signs

$$\varepsilon_k = \operatorname{sgn}(p^* - f)(x_k) = \sigma\,(-1)^k \qquad \text{for } k = 1, \ldots, n+1.$$

By these assumptions on φ, two conditions of the dual characterization (according to Theorem 3.48) are already satisfied, that is (a) $\|\varphi\|_\infty = 1$ and (b) $\varphi(p^* - f) = \|p^* - f\|_\infty$.

Now this problem is concerning condition (c) of the dual characterization in Theorem 3.48. To this end, consider using divided differences (cf. Definition 2.10) to construct, from given alternation points

$$a \le x_1 < \ldots < x_n \le b$$

and $\sigma \in \{\pm 1\}$, a coefficient vector $\lambda = (\lambda_1, \ldots, \lambda_{n+1})^T \in \Lambda_{n+1}$ satisfying

$$\varphi(p) = 0 \qquad \text{for all } p \in \mathcal{P}_{n-1}.$$

Exercise 5.49. Let $\mathcal{F} = \mathscr{C}[0, 2\pi]$ and $\mathcal{S} = \mathcal{P}_1$. Moreover, for $n \in \mathbb{N}$, let

$$f_n(x) = \sin(nx) \qquad \text{for } x \in [0, 2\pi].$$

(a) Compute the unique best approximation $s_n^* \in \mathcal{P}_1$ to f_n w.r.t. $\|\cdot\|_\infty$.
(b) How many alternation points occur for the error function $s_n^* - f_n$ in (a)? But should not there only be *three* alternation points?

Exercise 5.50. Let $\mathcal{F} = \mathscr{C}[-2,1]$ be equipped with the maximum norm $\|\cdot\|_\infty$. Compute the unique best approximation $p^* \in \mathcal{P}_2$ from $\mathcal{P}_2$ to the function $f \in \mathscr{C}[-2,1]$, defined as

$$f(x) = |x+1| \qquad \text{for } x \in [-2,1]$$

with respect to the maximum norm $\|\cdot\|_\infty$.

Moreover, determine the set of extremal points $X = E_{p^*-f}$, along with a constant $K > 0$ satisfying

$$\|p - f\|_{\infty,X} \ge \|p^* - f\|_\infty + K \cdot \|p - p^*\|_{\infty,X} \qquad \text{for all } p \in \mathcal{P}_2.$$

Plot the graphs of f and the best approximation p^* to f in one figure.

Exercise 5.51. Let $\mathcal{F} = \mathscr{C}[0,2]$ be equipped with the maximum norm $\|\cdot\|_\infty$. Determine the strongly unique best approximation $p^* \in \mathcal{P}_1$ from $\mathcal{P}_1$ to the function $f \in \mathscr{C}[0,2]$, defined as

$$f(x) = \exp\bigl(-(x-1)^2\bigr) \qquad \text{for } x \in [0,2]$$

with respect to the maximum norm $\|\cdot\|_\infty$.

Moreover, determine a constant $K > 0$ satisfying

$$\|p - f\|_\infty - \|p^* - f\|_\infty \geq K \cdot \|p - p^*\|_\infty \qquad \text{for all } p \in \mathcal{P}_1.$$

Use this inequality to conclude the uniqueness of the best approximation $p^* \in \mathcal{P}_1$ yet once more.

Exercise 5.52. Let $\mathcal{S} \subset \mathscr{C}[a,b]$ be a Haar space with $\dim(\mathcal{S}) = n+1 \in \mathbb{N}$.

Prove the *Haar condition*: If $s \in \mathcal{S} \setminus \{0\}$ has on the interval $[a,b]$ exactly m zeros from which k zeros are without sign change, then we have $m+k \leq n$.

Exercise 5.53. In this problem, let $I \subset \mathbb{R}$ be a compact set containing sufficiently many points, respectively. Analyze whether or not the following function systems $\mathcal{H} = (s_1, \ldots, s_n) \in (\mathscr{C}(I))^n$ are a Haar system on I.

(a) $\mathcal{H} = (x, 1/x)$ for $I \subset (0,\infty)$.
(b) $\mathcal{H} = (1/(x-c_0), 1/(x-c_1))$ for $I \subset \mathbb{R} \setminus \{c_0, c_1\}$, where $c_0 \neq c_1$.
(c) $\mathcal{H} = (1, x^2, x^4, \ldots, x^{2n})$ for $I = [-1,1]$.
(d) $\mathcal{H} = (1, x, \ldots, x^n, g(x))$ for a compact interval $I = [a,b]$, where $g \in \mathscr{C}^{n+1}[a,b]$ with $g^{(n+1)} \geq 0$ and $g^{(n+1)} \not\equiv 0$ on $[a,b]$.

Exercise 5.54. For $n \in \mathbb{N}_0$, let $\mathcal{T}_n^e$ be the linear space of all *even* real-valued trigonometric polynomials of degree at most n, and let $\mathcal{T}_n^o$ be the linear space of all *odd* real-valued trigonometric polynomials of degree at most n.

(a) Show that $\mathcal{T}_n^e$ is a Haar space on the interval $[0,\pi)$.
(b) Determine the dimension of $\mathcal{T}_n^e$.
(c) Is $\mathcal{T}_n^o$ a Haar space on the interval $[0,\pi)$?
(d) Is $\mathcal{T}_n^o$ a Haar space on the open interval $(0,\pi)$?
(e) Determine the dimension of $\mathcal{T}_n^o$.

Exercise 5.55. Prove the following results.

(a) The functions

$$s_0(x) = 1, \quad s_1(x) = x\cos(x), \quad s_2(x) = x\sin(x)$$

are a Haar system on $[0,\pi]$.

(b) There is *no* two-dimensional subspace of

$$\mathcal{S} = \operatorname{span}\{s_0, s_1, s_2\} \subset \mathscr{C}[0,\pi],$$

which is a Haar space on $[0,\pi]$.

Exercise 5.56. For $n \in \mathbb{N}_0$, let $\mathcal{S} \subset \mathscr{C}[a,b]$ be a $(n+1)$-dimensional linear subspace of $\mathscr{C}[a,b]$. Moreover, let $\mathcal{S}$ satisfy the *weak Haar condition* on $[a,b]$, according to which any $s \in \mathcal{S}$ has at most n sign changes in $[a,b]$.

Prove the following statements for $f \in \mathscr{C}[a,b]$.

(a) If there is an alternation set for $s \in \mathcal{S}$ and f of length $n+2$, so that there are $n+2$ pairwise distinct alternation points $a \leq x_0 < \ldots < x_{n+1} \leq b$ and one sign $\sigma \in \{\pm 1\}$ satisfying

$$(s-f)(x_k) = \sigma\,(-1)^k\,\|s-f\|_\infty \qquad \text{for all } k = 0, \ldots, n+1,$$

then s is a best approximation to f from $\mathcal{S}$ with respect to $\|\cdot\|_\infty$.

(b) The converse of statement (a) is false (for the general case).

Exercise 5.57. Let $\mathcal{F} = \mathscr{C}[a,b]$ and $\mathcal{S} \subset \mathcal{F}$ be a Haar space on $[a,b]$ of dimension $n+1$ containing the constant functions. Moreover, let $f \in \mathcal{F} \setminus \mathcal{S}$, such that

$$\operatorname{span}\{\mathcal{S} \cup \{f\}\}$$

is a Haar space on $[a,b]$. Finally, $s^* \in \mathcal{S}$ be the unique best approximation to f from $\mathcal{S}$ with respect to $\|\cdot\|_\infty$.

Show that the error function $f - s^*$ has exactly $n+2$ extremal points

$$a = x_0 < \ldots < x_{n+1} = b,$$

where $f - s^*$ is strictly monotone between neighbouring extremal points.

Exercise 5.58. In this programming exercise, we wish to compute for any $n \in \mathbb{N}$ the strongly unique best approximation $p^* \in \mathcal{P}_{n-1}$ to $f \in \mathscr{C}[a,b] \setminus \mathcal{P}_{n-1}$ from $\mathcal{P}_{n-1}$ w.r.t. $\|\cdot\|_{\infty,X}$ on a point set $X = (x_1, \ldots, x_{n+1}) \in [a,b]^{n+1}$, so that

$$\|p^* - f\|_{\infty,X} < \|p - f\|_{\infty,X} \qquad \text{for all } p \in \mathcal{P}_{n-1} \setminus \{p^*\}.$$

To this end, implement a function called `mybestpoly` with header

```
[alpha,eta] = mybestpoly(f,X),
```

which returns on input point set X (of length $|X| = n+1$) the Newton coefficients $\alpha = (\alpha_0, \ldots, \alpha_{n-1}) \in \mathbb{R}^n$ of the best approximation

$$p^*(x) = \sum_{k=0}^{n-1} \alpha_k\, \omega_k(x) \quad \text{where } \omega_k(x) = \prod_{j=1}^{k} (x - x_j) \in \mathcal{P}_k \text{ for } 0 \leq k \leq n-1$$

to f w.r.t. $\|\cdot\|_{\infty,X}$, along with the minimal distance $\eta_X = \|f - s^*\|_{\infty,X}$.

Exercise 5.59. To *efficiently* evaluate the best approximation $p^* \in \mathcal{P}_{n-1}$ from Exercise 5.58, we use the *Horner*[10] *scheme* (a standard numerical method, see e.g. [28, Section 5.3.3].

[10] William George Horner (1786-1837), English mathematician

To this end, implement a function called `mynewtonhorner` with header

`[p] = mynewtonhorner(X,alpha,x),`

which returns on an input point set $X = \{x_1, \dots, x_{n+1}\} \subset [a,b]$, Newton coefficients $\alpha = (\alpha_0, \dots, \alpha_{n-1}) \in \mathbb{R}^n$ and $x \in \mathbb{R}$ the value

$$p(x) = \sum_{k=0}^{n-1} \alpha_k \, \omega_k(x) \in \mathcal{P}_{n-1},$$

where the evaluation of p at x should rely on the Horner scheme.

Exercise 5.60. Implement the Remez exchange, Algorithm 8. To this end, write a function called `myremezexchange` with header

`[X] = myremezexchange(X,epsilon,x),`

which returns, on input reference set $X = (x_1, \dots, x_{n+1}) \in [a,b]^{n+1}$, an extremal point $x = x^* \in [a,b] \setminus X$ satisfying

$$|(p^* - f)(x^*)| = \|p^* - f\|_\infty$$

and a sign vector $\varepsilon = (\varepsilon_1, \varepsilon_2) \in \{\pm 1\}^2$ satisfying

$$\varepsilon_1 = \operatorname{sgn}(p^* - f)(x_1) \quad \text{and} \quad \varepsilon_2 = \operatorname{sgn}(p^* - f)(x^*)$$

the updated reference set X_+ (as output by Algorithm 8), i.e.,

$$X_+ = (X \setminus \{x_j\}) \cup \{x^*\} \quad \text{for one } 1 \le j \le n+1.$$

Exercise 5.61. Implement the Remez algorithm, Algorithm 9. To this end, write a function `myremez` with header

`[alpha,eta,X,its] = myremez(f,X),`

which returns, on input function $f \in \mathscr{C}[a,b] \setminus \mathcal{P}_{n-1}$ and an initial reference set $X = (x_1, \dots, x_{n+1}) \in [a,b]^{n+1}$, the Newton coefficients $\alpha = (\alpha_0, \dots, \alpha_{n-1})$ of the (strongly unique) best approximation $p^* \in \mathcal{P}_{n-1}$ to f from $\mathcal{P}_{n-1}$ w.r.t. $\|\cdot\|_\infty$, the minimal distance $\eta = \|p^* - f\|_\infty$, a set of alternation points $X \subset E_{p^*-f}$, and the number `its` of the performed Remez iterations. For your implementation, use the functions `mybestpoly` (from Exercise 5.58), `mynewtonhorner` (Exercise 5.59) and `myremezexchange` (Exercise 5.60).

Verify your function `myremez` by using the following examples.

(a) $f(x) = \sqrt[3]{x}$, $[a,b] = [0,1]$, $X = \left(0, \frac{1}{2}, \frac{3}{4}, 1\right)$;
(b) $f(x) = \sin(5x) + \cos(6x)$, $[a,b] = [0,\pi]$, $X = \left(0, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \pi\right)$.

Exercise 5.62. Analyze for the case $\mathcal{S} = \mathcal{P}_{n-1}$ the asymptotic computational complexity for only *one* iteration of the Remez algorithm, Algorithm 9.

(a) Determine the costs for the minimal distance $\eta_k = \|s_k^* - f\|_{\infty,X_k}$.
Hint: Use divided differences (according to Proposition 5.35).
(b) Determine the costs for computing the Newton coefficients of s_k^*.
Hint: Reuse the divided differences from (a).
(c) Sum up the required asymptotic costs in (a) and (b).

How do you *efficiently* compute the update η_{k+1} from information that is required to compute η_k?

Exercise 5.63. Assuming the notations of the *Remez algorithm*, Algorithm 9, we consider the (global) distance

$$\rho_k = \|s_k^* - f\|_\infty \qquad \text{for } k \in \mathbb{N}_0$$

between $f \in \mathscr{C}[a,b]$ and the current best approximation $s_k^* \in \mathcal{S}$ to f, for the current reference set $X_k = (x_1^{(k)}, \ldots, x_{n+1}^{(k)}) \in [a,b]^{n+1}$ and w.r.t. $\|\cdot\|_{\infty,X_k}$.

Show that the sequence $(\rho_k)_{k\in\mathbb{N}_0}$ is *not necessarily* strictly increasing. To this end, construct a simple (but non-trivial) counterexample.